

Customer Segmentation Using K-Means Clustering

1. Introduction

Customer segmentation is the process of dividing customers into groups based on similar characteristics and behavior. In this project, customer data was analyzed to identify different groups based on age, annual income, and spending score. K-Means clustering was used to create the customer segments and understand their characteristics.

2. Dataset Selection

The **Mall Customers Dataset** was used for this project. It contains information about 200 customers, including gender, age, annual income, and spending score.

The main features used in the analysis were:

- Age
- Annual Income (k\$)
- Spending Score (1-100)

CustomerID was treated only as an identifier and was not used for clustering.

3. Data Preprocessing

The dataset was first checked for missing values, duplicate records, data types, and its overall structure. No missing-value treatment was required.

Gender was kept for exploratory analysis, while CustomerID was excluded from the clustering features because it does not describe customer behavior.

The three numerical features — Age, Annual Income, and Spending Score — were selected for clustering. Since K-Means is distance-based, **StandardScaler** was used to bring the features to a comparable scale.

4. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the basic characteristics of the customers.

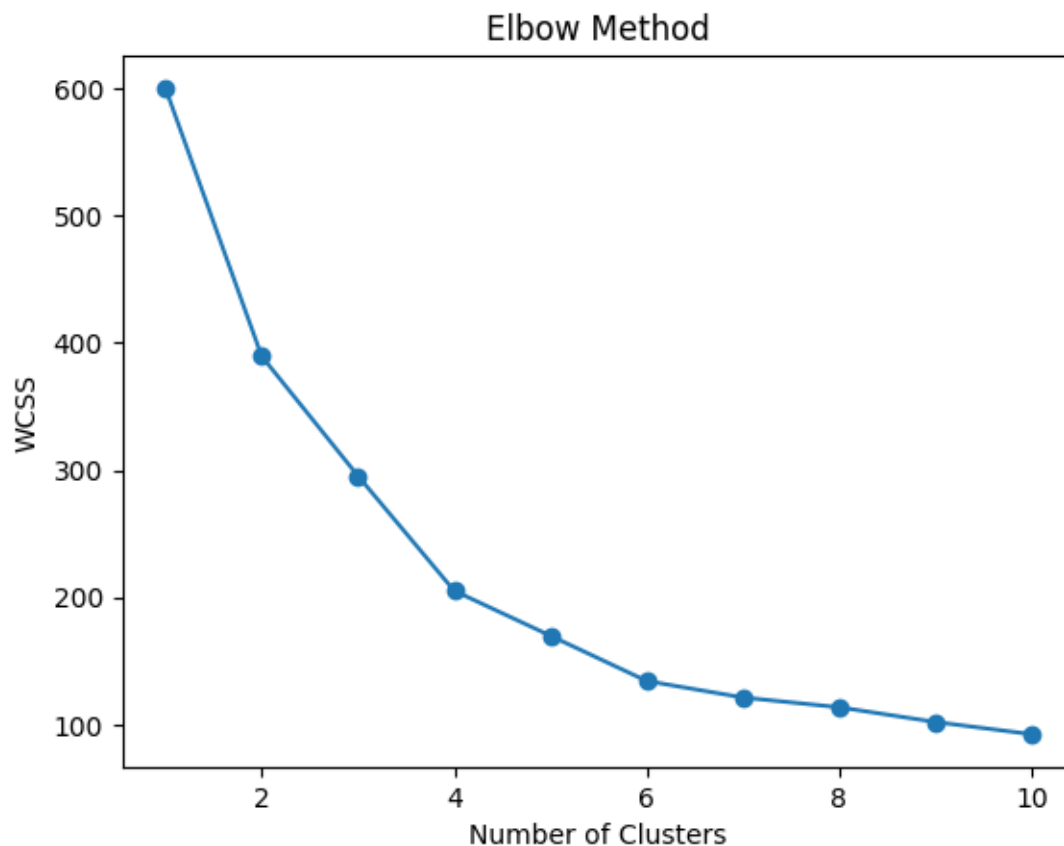
Three graphs were created:

- **Gender Distribution:** Female customers were slightly more represented than male customers.
- **Age Distribution:** Customers covered a broad age range, approximately from 20 to 70 years.
- **Spending Score Distribution:** A large number of customers had spending scores between approximately 40 and 60.

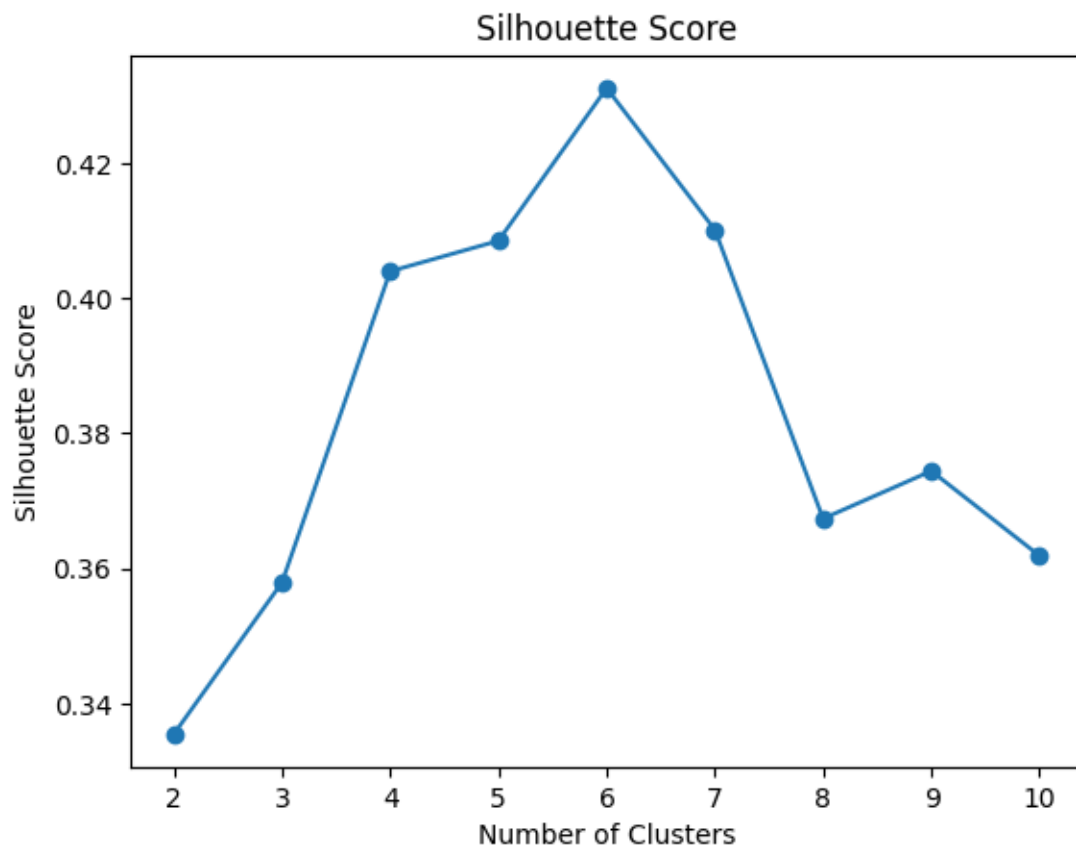
These observations provided an initial understanding of the customer data before clustering.

5. Determining the Number of Clusters

The **Elbow Method** was used to examine different numbers of clusters by comparing the WCSS values. The curve showed a noticeable change around 4–5 clusters.



The **Silhouette Score** was then used to further evaluate the cluster quality. The highest score was obtained for **6 clusters**.



Based on these results, 6 clusters were selected for the final K-Means model.

6. K-Means Clustering

K-Means clustering was applied using the scaled values of Age, Annual Income, and Spending Score.

The final model divided the 200 customers into **six segments** with different demographic and spending characteristics.

The identified segments were:

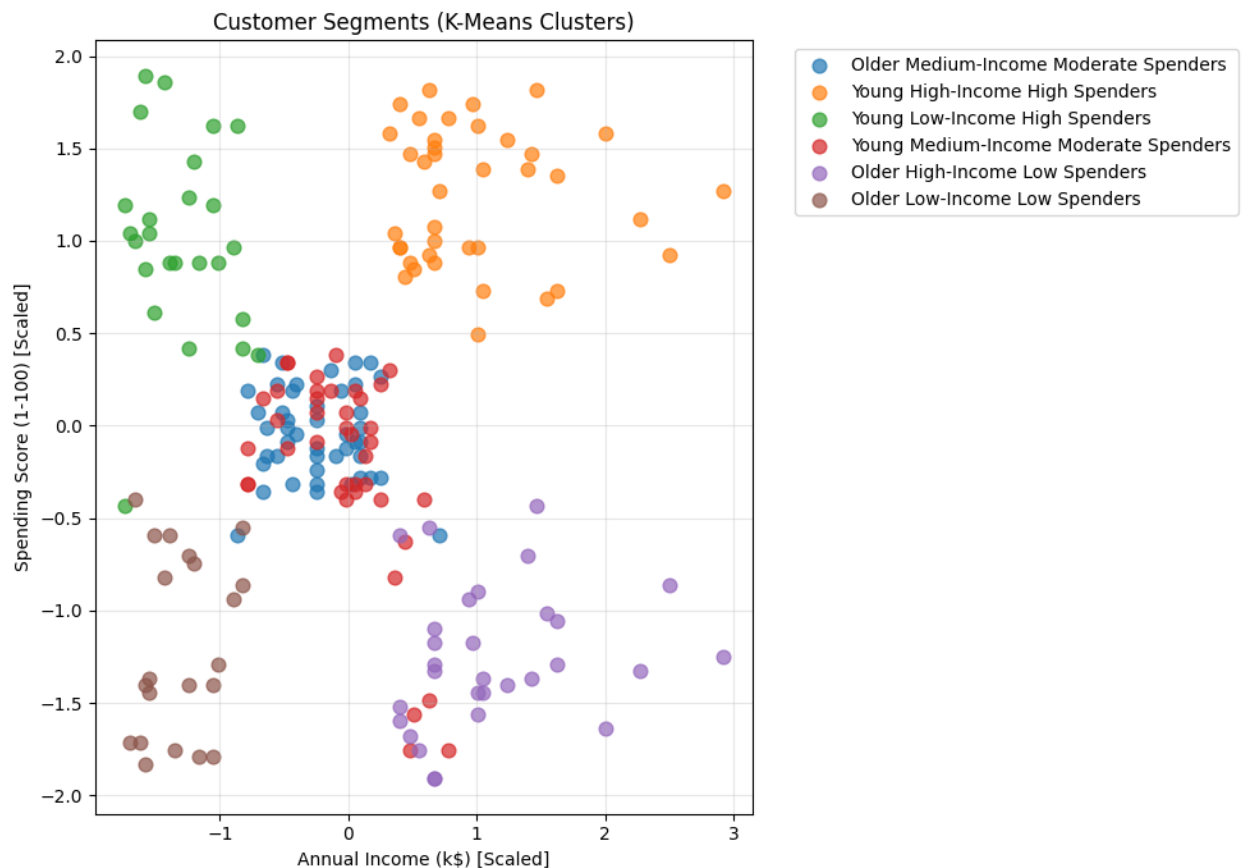
1. Older High-Income Low Spenders
2. Older Low-Income Low Spenders
3. Older Medium-Income Moderate Spenders

4. Young High-Income High Spenders
5. Young Low-Income High Spenders
6. Young Medium-Income Moderate Spenders

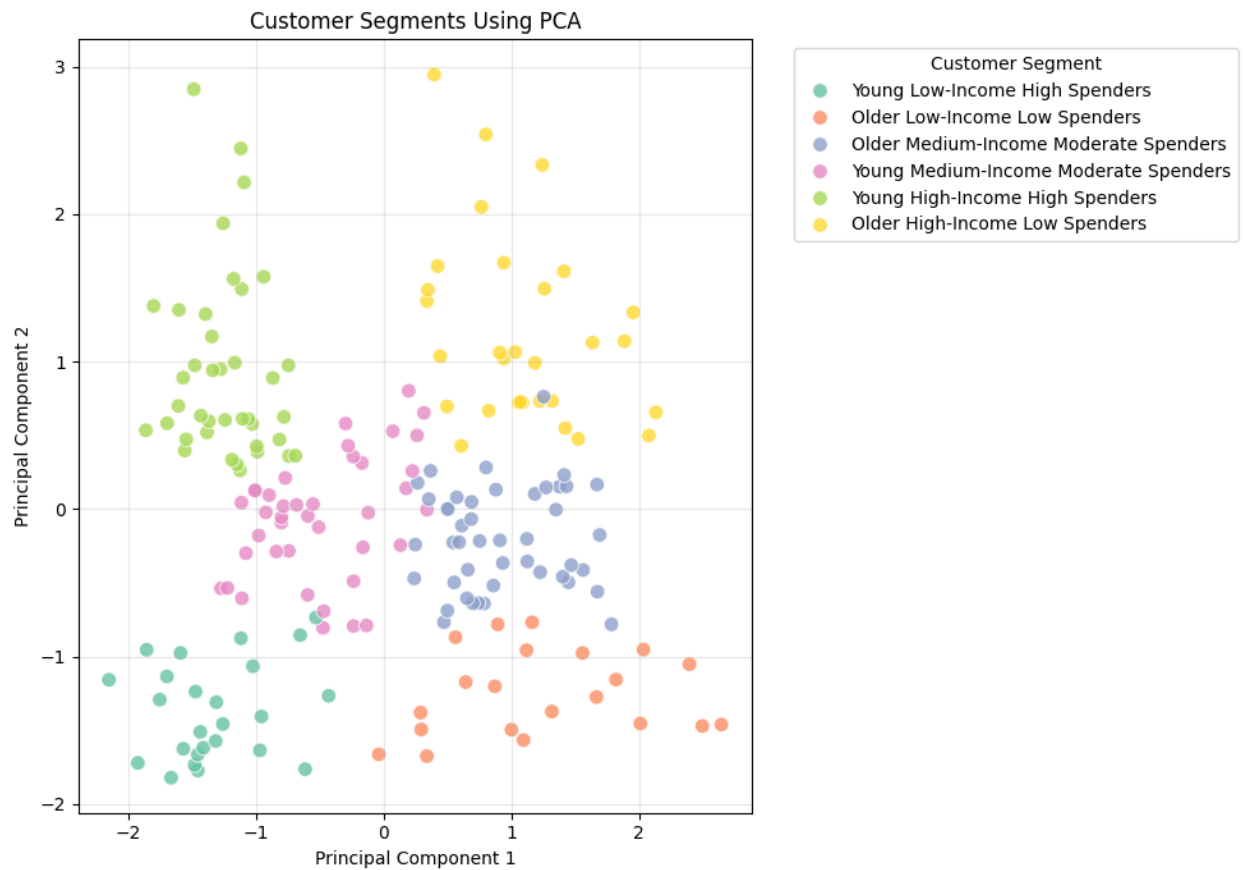
7. Cluster Visualization

Several visualizations were used to understand the resulting clusters.

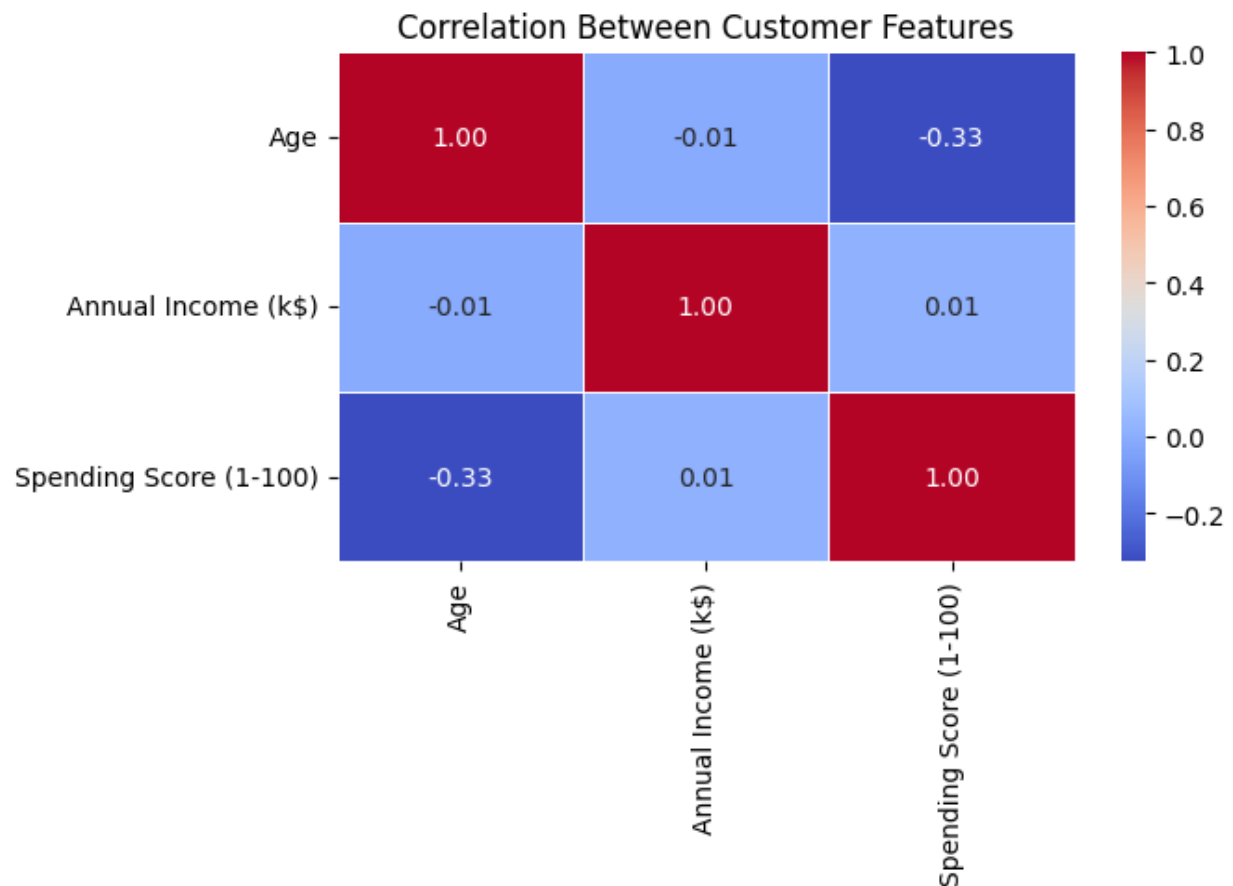
A scatter plot was created using Annual Income and Spending Score to show the distribution of customers across the six clusters.



PCA was also applied to reduce the three-dimensional feature space to two dimensions. The PCA scatter plot provided another view of the separation between the customer segments.



A correlation heatmap was used to examine the relationships between Age, Annual Income, and Spending Score.



8. Segment Interpretation and Business Insights

The clusters showed different combinations of age, income, and spending behavior.

- **High-Income High Spenders:** Premium products and loyalty programs can be considered.
- **Low-Income High Spenders:** Discounts and affordable offers may be useful.
- **Moderate Spenders:** Personalized recommendations and promotions can be used.
- **High-Income Low Spenders:** Targeted engagement and incentives may help increase engagement.
- **Low-Income Low Spenders:** Value-based and budget-friendly offers can be considered.
- **Moderate-Income Customers:** Retention campaigns and personalized offers can be used.

These segments can help businesses understand their customers and plan more targeted marketing and engagement strategies.

9. Conclusion

K-Means clustering divided the customers into **six segments** based on age, income, and spending behavior. The analysis showed clear differences between customers with high, moderate, and low spending patterns.

Overall, customer segmentation provides useful information for personalized offers, customer engagement, retention, and marketing decisions.